proper fills during fabrication by placing geometry on relevant layers

Routing. Electric comes with a number of routing modules. The maze router helps in connecting single wires between points, while the sea-of-gates router uses multi-threading to run wires faster. There are two other routers that ensure wires are properly placed. Other than that, the auto router creates exclusive connections in overlapping regions. The mimic router adds extra wires in situations similar to those when the user runs by hand. The river router connects parallel wires in a channel between cells.

Network consistency checking. Electric adopts graph isomorphism to compare a layout with its equivalent schematic and evaluates the network consistency. It is also able to compare two different versions of a schematic or layout for network consistency checking.

Logical Effort. Logical Effort helps in marking digital logic gates with fan-out information, ensuring optimal speed of the circuit.

VHDL. The VHSIC Hardware Description Language (VHDL) system uses a layout to generate VHDL, and compiles the program to netlists of various formats. Then, with the help of built-in simulators, these netlists can be simulated and converted into a layout with the silicon compiler, or saved to memory.

Silicon compiler. The netlist created from VHDL is fed to the silicon compiler for placing and routing standard cells.

Compaction. The compactor tool ensures minimal spacing, or compaction, of the circuit by adjusting the geometry along X and Y axes.

Supported file formats

Electric has its own specific format of reading and writing circuitry files. Yet, it supports an exhaustive array of file formats to match compatibility with other EDA tools. File formats include Caltech Intermediate Format (CIF), Calma Interchange Format (GDS II), Electronic Design Interchange Format (EDIF), LEF and DEF formats, DXF autoCAD format, VHDL and Verilog languages, schematic capture packages like EAGLE, PADS, EDAD and SUE, and finally, printing formats like PNG, PostScript and SVG.

In a nutshell

Electric is an open-source GNU project undertaking. Coded in JAVA, this software can run on majority of platforms. Electric is a simple and compact EDA tool with a large repertoire of facilities that can be put to best use. EFY